Low-frequency phase including spring dynamics

New fit begins at 3 Hz; old model below 25 Hz is extrapolation. Individual channel delays are included.
Each curve is unwrapped independently; only one constant whole-curve 360-deg branch shift is allowed. No smoothing.

Validation LRM — — Old model (below 25 Hz: extrapolation) ——— Joint low-frequency model

$u_1 \to y_1$ $u_2 \to y_1$ $u_3 \to y_1$ $u_4 \to y_1$

$u_1 \to y_2$ $u_2 \to y_2$ $u_3 \to y_2$ $u_4 \to y_2$

$u_1 \to y_3$ $u_2 \to y_3$ $u_3 \to y_3$ $u_4 \to y_3$

$u_1 \to y_4$ $u_2 \to y_4$ $u_3 \to y_4$ $u_4 \to y_4$

Unwrapped phase (deg)

Frequency (Hz)